重要声明

本文档仅作为学习思路、代码原理参考资料。
严禁直接复制全文提交作为个人实验报告，抄袭会造成课程成绩失效、学术不端处分。
请自主理解原理、修改代码与行文，独立完成实验报告。

哈尔滨工业大学（深圳）

电磁场实验报告

静电场有限差分法编程实现

课程名称： 电磁场

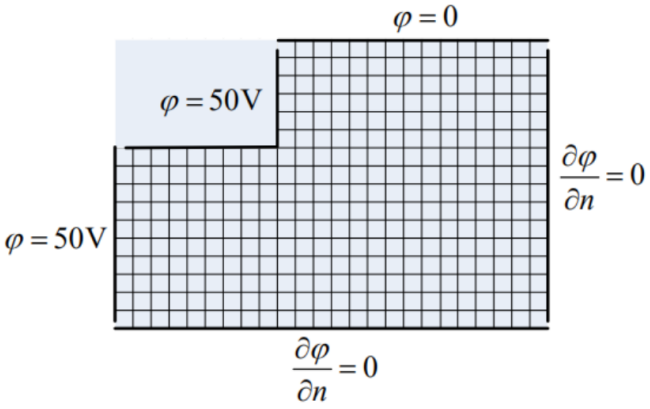
实验名称： 实验一有限差分法求解二维静电场

完成日期： 2026 年 05 月 09 日

Harbin Institute of Technology, Shenzhen

1 实验题图

尾数 6：试用高斯-塞德尔迭代法和超松弛迭代法确定二维静电场域的电位分布。



2 实验程序

```
% ================================
% 二维静电场求解：SOR迭代 + 高斯-赛德尔迭代 尾号06
% ================================
clear; clc; close all;

% 公共参数
hx=25;hy=17; %设置网格节点数
m=24;n=16;%横纵向网格数

%% ====================== 一、高斯-赛德尔迭代法 ======================
disp('====================␣高斯-赛德尔迭代结果␣====================');
v_gs=ones(hy,hx);
v_gs(1:6,17:25)=zeros(6,9);
% Dirichlet 边界条件
v_gs(1,10:25)=0;
v_gs(2:7,10)=50;
v_gs(7,1:10)=50;
v_gs(7:16,1)=50;

v2_gs = v_gs;
maxt=1;
k_gs=0;

while(maxt>1e-6)
    k_gs=k_gs+1;
    maxt=0;
    v_old = v2_gs;

    % 上区域迭代
    for i=2:7
```

```
        for j=11:24
            v2_gs(i,j) = (v2_gs(i,j+1)+v2_gs(i+1,j)+v2_gs(i-1,j)+v2_gs(i,j-1))/4;
            t=abs(v2_gs(i,j)-v_old(i,j));
            if(t>maxt) maxt=t;end
        end
    end
    % 下区域迭代
    for i=8:(hy-1)
        for j=2:(hx-1)
            v2_gs(i,j) = (v2_gs(i,j+1)+v2_gs(i+1,j)+v2_gs(i-1,j)+v2_gs(i,j-1))/4;
            t=abs(v2_gs(i,j)-v_old(i,j));
            if(t>maxt) maxt=t;end
        end
    end

    % Neumann 边界条件：右、下偏导为0
    v2_gs(7:hy-1,hx)=v2_gs(7:hy-1,hx-1);
    v2_gs(hy,1:hx)=v2_gs(hy-1,1:hx);
end

% 输出高斯-赛德尔结果
fprintf('高斯-赛德尔迭代次数：%d\n',k_gs);
disp('高斯-赛德尔最终电位矩阵：');
disp(v2_gs);
v_gs_plot = v2_gs(hy:-1:1,:);

%% ====================== 二、SOR超松弛迭代法 ======================
disp('====================␣SOR超松弛迭代结果␣====================');
v1=ones(hy,hx);
v1(1:6,17:25)=zeros(6,9);
%边界的 Dirichlet 边界条件值
v1(1,10:25)=ones(1,16)*0;
v1(2:7,10)=ones(6,1)*50;
v1(7,1:10)=ones(1,10)*50;
v1(7:16,1)=50;

%计算松弛因子
t1=(cos(pi/m)+cos(pi/n))/2;
w=2/(1+sqrt(1-t1*t1));
v2=v1;maxt=1;t=0;%初始化
k=0;
while(maxt>1e-6)%由 v1 迭代，算出 v2，迭代精度为 0.000001
    k=k+1;%计算迭代次数
    maxt=0;
    for i=2:7%从 2 到 7 行循环
        for j=11:24%从 2 到 15 列循环
            v2(i,j)=v1(i,j)+(v1(i,j+1)+v1(i+1,j)+v2(i-1,j)+v2(i,j-1)-4*v1(i,j))*w
                /4;%拉普拉斯方程差分式
            t=abs(v2(i,j)-v1(i,j));
            if(t>maxt) maxt=t;end
```

```
        end
    end
    for i=8:(hy-1)%从 8 到 hy-1 行循环
        for j=2:(hx-1)%从 2 到 hx-1 列循环
            v2(i,j)=v1(i,j)+(v1(i,j+1)+v1(i+1,j)+v2(i-1,j)+v2(i,j-1)-4*v1(i,j))*w
                /4;%拉普拉斯方程差分式
            t=abs(v2(i,j)-v1(i,j));
            if(t>maxt) maxt=t;end
        end
    end
    % 最右侧边界 偏导为0 Neumann边界条件
    v2(7:hy-1,hx)=v2(7:hy-1,hx-1);
    % 最下侧边界 偏导为0 Neumann边界条件
    v2(hy,1:hx)=v2(hy-1,1:hx);
    v1=v2;
end
v1=v2(hy:-1:1,:);
% 输出迭代收敛后的最终电位矩阵
fprintf('SOR迭代次数: %d\n',k);
disp('SOR迭代最终电位矩阵: ');
disp(v2);

%% ====================== 三、统一绘图对比 ======================
figure('Position',[100,100,1000,800]);

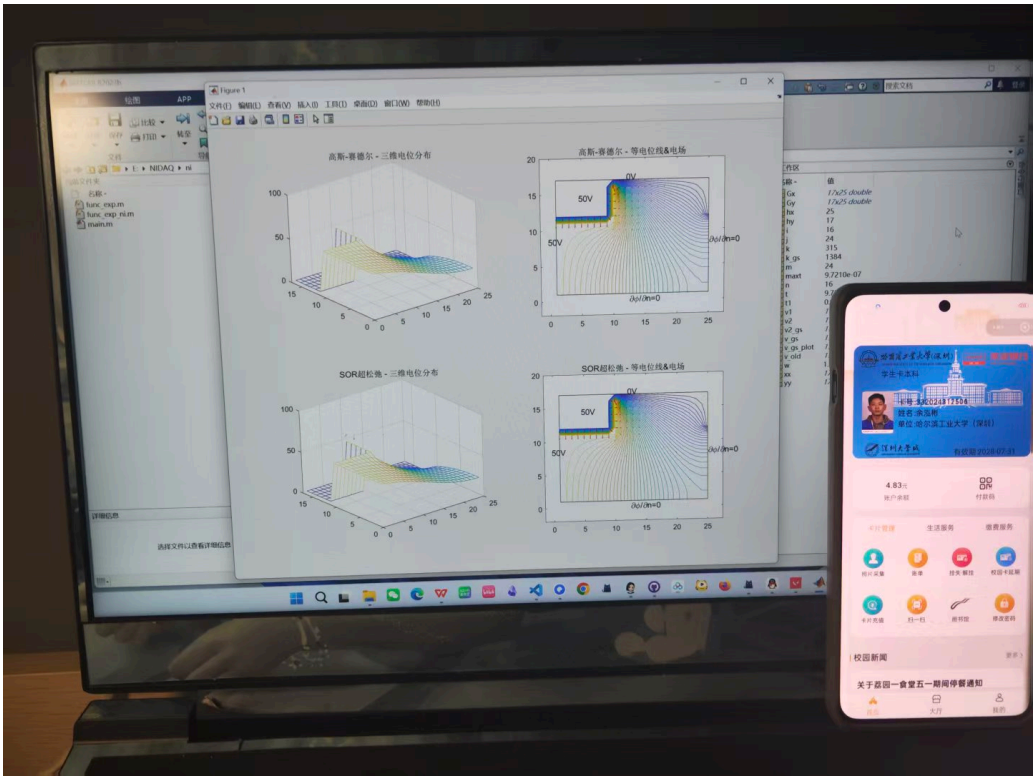
% 上排: 高斯-赛德尔
subplot(2,2,1),mesh(v_gs_plot)
axis([0,25,0,17,0,100])
title('高斯-赛德尔␣-␣三维电位分布')
subplot(2,2,2),contour(v_gs_plot,50)
hold on
[xx,yy]=meshgrid(1:hx,1:hy);
[Gx,Gy]=gradient(v_gs_plot,0.6,0.6);
quiver(xx,yy,Gx,Gy,'r')
axis([-1.5,hx+2.5,-2,20])
plot([1,1,hx,hx,1],[1,hy,hy,1,1],'k')
text(hx/2-0.5,hy+0.4,'0V','fontsize',11);
text(hx/2,0.3,'\partial\phi/\partialn=0','fontsize',11);
text(-0.3,hy/2,'50V','fontsize',11);
text(hx+0.1,hy/2,'\partial\phi/\partialn=0','fontsize',11);
text(hx/2-8,hy/2+6,'50V','fontsize',11);
title('高斯-赛德尔␣-␣等电位线&电场')
hold off

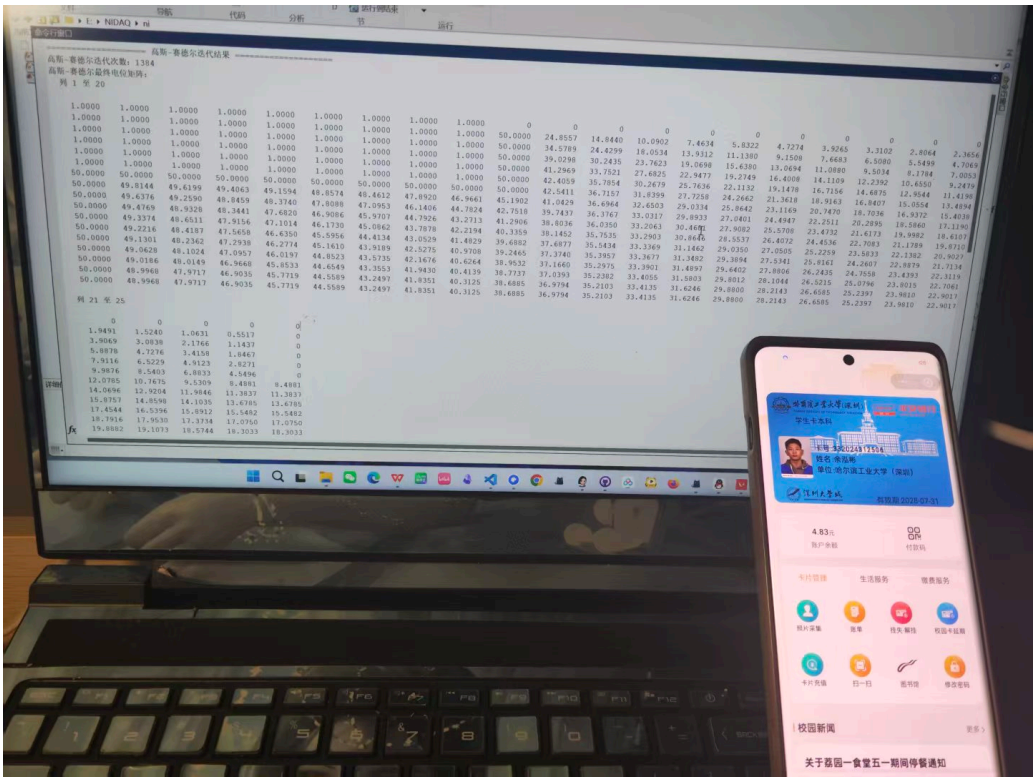
% 下排: SOR迭代
subplot(2,2,3),mesh(v1)
axis([0,25,0,17,0,100])
title('SOR超松弛␣-␣三维电位分布')
subplot(2,2,4),contour(v1,50)
hold on
```

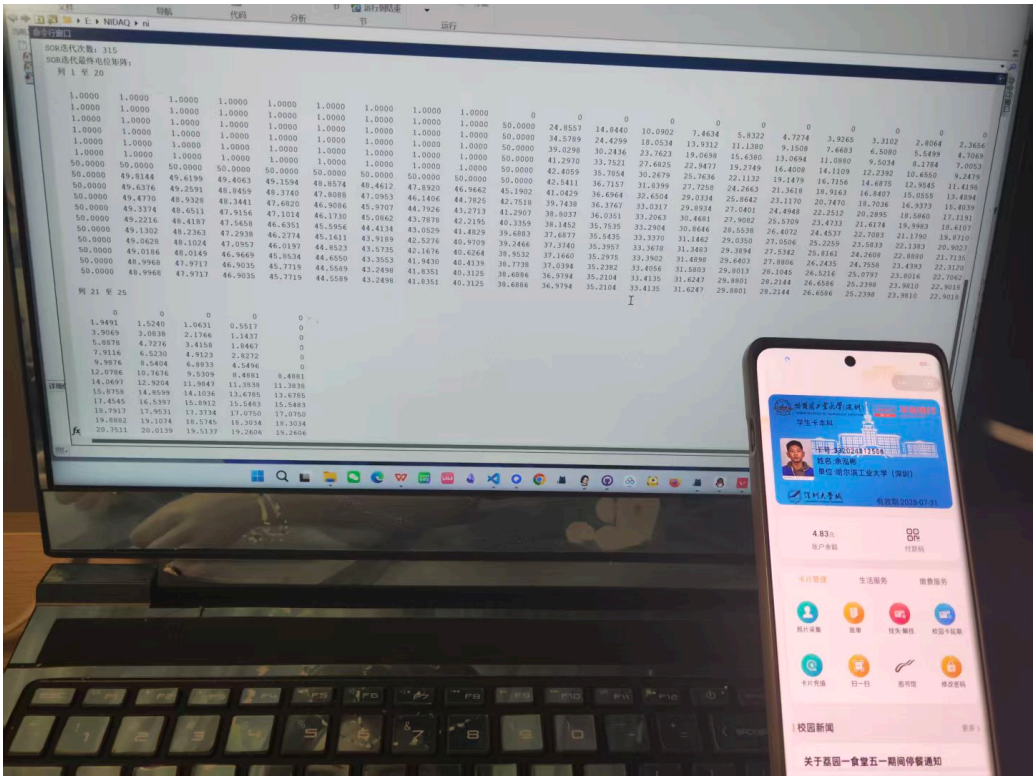
```
[Gx,Gy]=gradient(v1,0.6,0.6);
quiver(xx,yy,Gx,Gy,'r')
axis([-1.5,hx+2.5,-2,20])
plot([1,1,hx,hx,1],[1,hy,hy,1,1],'k')
text(hx/2-0.5,hy+0.4,'0V','fontsize',11);
text(hx/2,0.3,'\partial\phi/\partialn=0','fontsize',11);
text(-0.3,hy/2,'50V','fontsize',11);
text(hx+0.1,hy/2,'\partial\phi/\partialn=0','fontsize',11);
text(hx/2-8,hy/2+6,'50V','fontsize',11);
title('SOR超松弛␣-␣等电位线&电场')
hold off
```

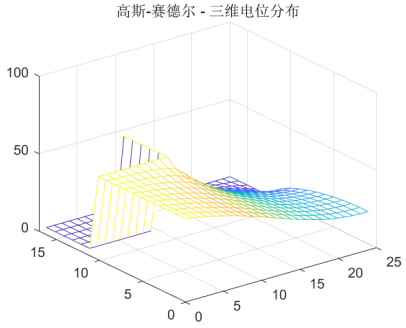
Listing 1: MATLAB 求解二维静电场有限差分迭代程序

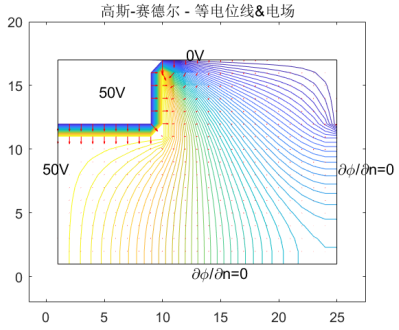
3 实验结果

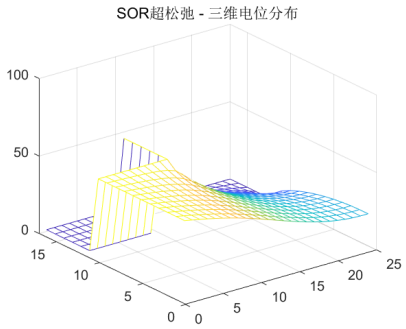


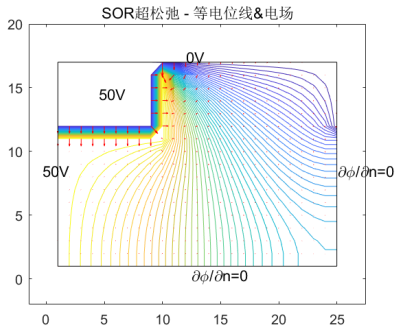




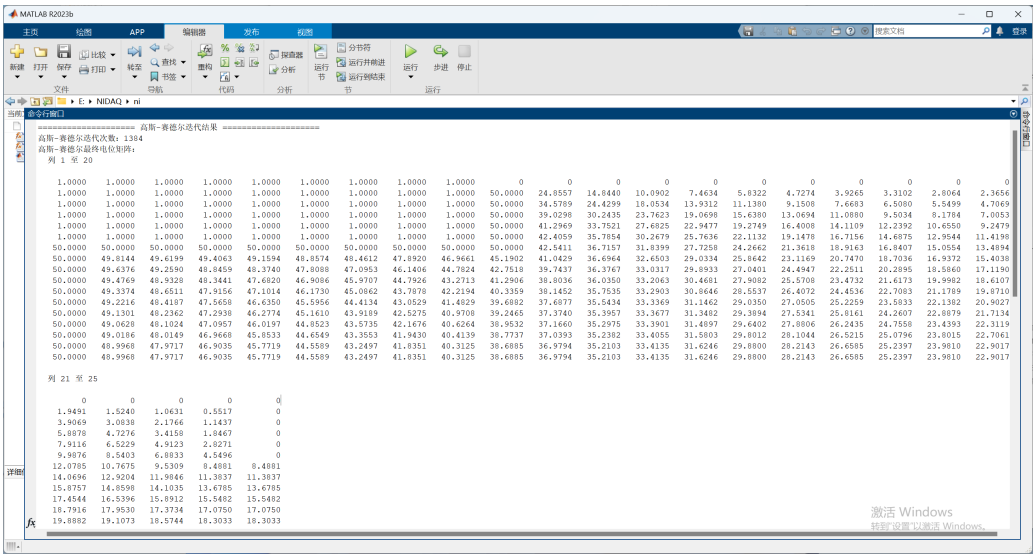


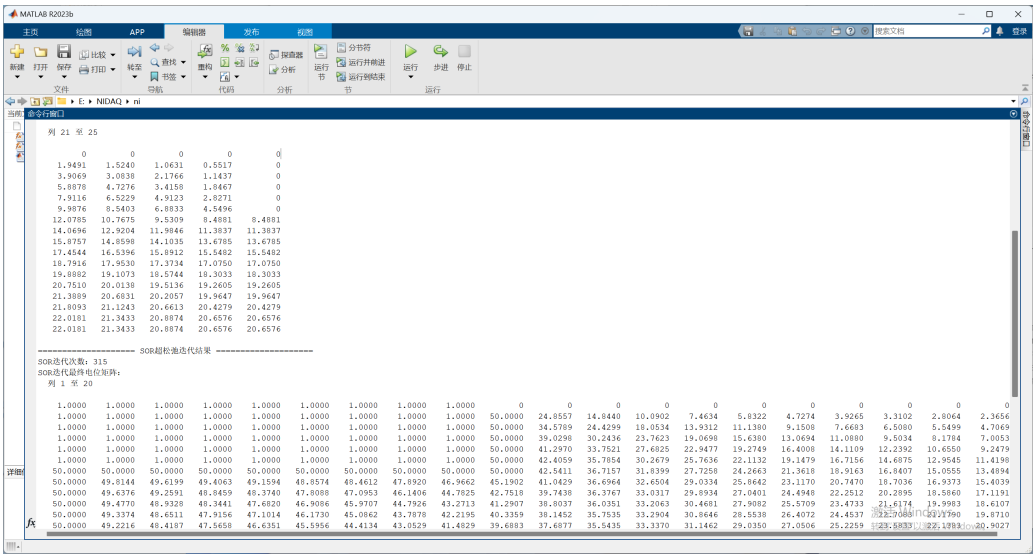


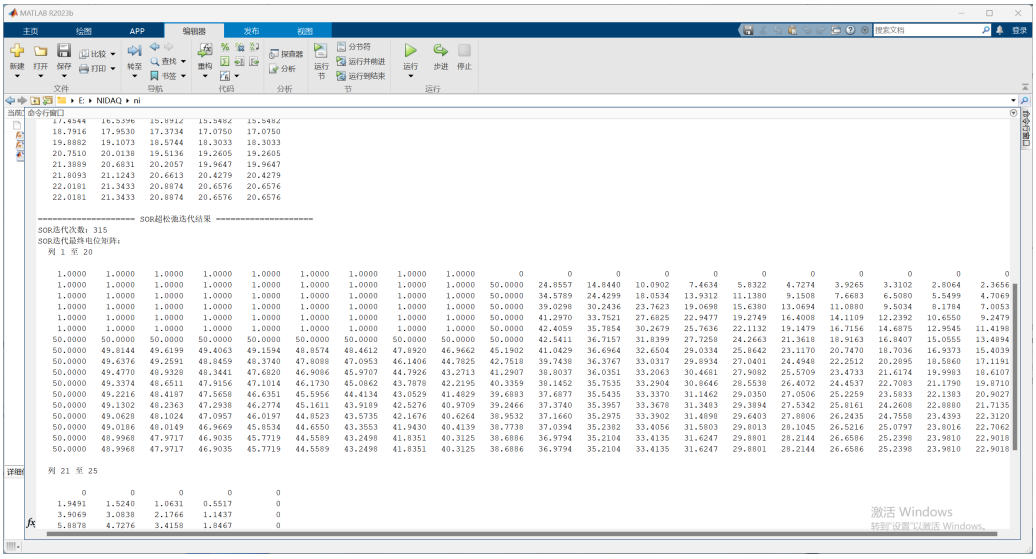


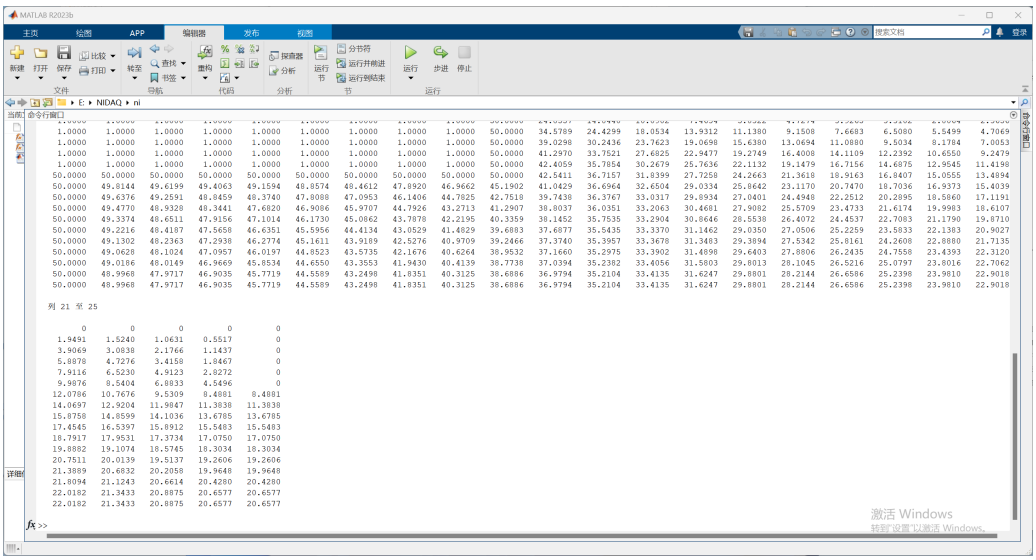


3.1 迭代输出与电位矩阵数据




MATLAB R2023b
E: ▸ NIDAQ ▸ ni
命令行窗口
列 21 至 25
0 0 0 0 0
1.9491 1.5240 1.0631 0.5517 0
3.9069 3.0838 2.1766 1.1437 0
5.8878 4.7276 3.4158 1.8467 0
7.9116 6.5229 4.9123 2.8271 0
9.9876 8.5403 6.8833 4.5496 0
12.0785 10.7675 9.5309 8.4881 8.4881
14.0696 12.9204 11.9846 11.3837 11.3837
15.8757 14.8598 14.1035 13.6785 13.6785
17.4544 16.5396 15.8912 15.5482 15.5482
18.7916 17.9530 17.3734 17.0750 17.0750
19.8882 19.1073 18.5744 18.3033 18.3033
20.7510 20.0138 19.5136 19.2605 19.2605
21.3889 20.6831 20.2057 19.9647 19.9647
21.8093 21.1243 20.6613 20.4279 20.4279
22.0181 21.3433 20.8874 20.6576 20.6576
22.0181 21.3433 20.8874 20.6576 20.6576
SOR超松弛迭代结果
SOR迭代次数: 315
SOR迭代最终电位矩阵:
列 1 至 20
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 0 0 0 0 0 0 0 0 0 0 0
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 50.0000 24.8557 14.8440 10.0902 7.4634 5.8322 4.7274 3.9265 3.3102 2.8064 2.3656
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 50.0000 34.5789 24.4299 18.0534 13.9312 11.1380 9.1508 7.6683 6.5080 5.5499 4.7069
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 50.0000 39.0298 30.2436 23.7623 19.0698 15.6380 13.0694 11.0880 9.5034 8.1784 7.0053
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 50.0000 41.2970 33.7521 27.6825 22.9477 19.2749 16.4008 14.1109 12.2392 10.6550 9.2479
1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 50.0000 42.4059 35.7854 30.2679 25.7636 22.1132 19.1479 16.7156 14.6875 12.9545 11.4198
50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 50.0000 42.5411 36.7157 31.8399 27.7258 24.2663 21.3618 18.9163 16.8407 15.0555 13.4894
50.0000 49.8144 49.6199 49.4063 49.1594 48.8574 48.4612 47.8920 46.9662 45.1902 41.0429 36.6964 32.6504 29.0334 25.8642 23.1170 20.7470 18.7036 16.9373 15.4039
50.0000 49.6376 49.2591 48.8459 48.3740 47.8088 47.0953 46.1406 44.7825 42.7518 39.7438 36.3767 33.0317 29.8934 27.0401 24.4948 22.2512 20.2895 18.5860 17.1191
50.0000 49.4770 48.9328 48.3441 47.6820 46.9086 45.9707 44.7926 43.2713 41.2907 38.8037 36.0351 33.2063 30.4681 27.9082 25.5709 23.4733 21.6174 19.9983 18.6107
50.0000 49.3374 48.6511 47.9156 47.1014 46.1730 45.0862 43.7878 42.2195 40.3359 38.1452 35.7535 33.2904 30.8646 28.5538 26.4072 24.4537 22.7083 21.1790 19.8710
50.0000 49.2216 48.4187 47.5658 46.6351 45.5956 44.4134 43.0529 41.4829 39.6883 37.6877 35.5435 33.3370 31.1462 29.0350 27.0506 25.2259 23.5833 22.1303 20.9027
fx
激活 Windows
转到"设置"以激活 Windows。


MATLAB R2023b
E: ▸ NIDAQ ▸ ni
命令行窗口
SOR超松弛迭代结果
SOR迭代次数: 315
SOR迭代最终电位矩阵:
列 1 至 20
列 21 至 25
0 0 0 0 0
1.9491 1.5240 1.0631 0.5517 0
3.9069 3.0838 2.1766 1.1437 0
5.8878 4.7276 3.4158 1.8467 0
fx
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E: ▸ NIDAQ ▸ ni
命令行窗口
列 21 至 25
0 0 0 0 0
1.9491 1.5240 1.0631 0.5517 0
3.9069 3.0838 2.1766 1.1437 0
5.8878 4.7276 3.4158 1.8467 0
7.9116 6.5230 4.9123 2.8272 0
9.9876 8.5404 6.8833 4.5496 0
12.0786 10.7676 9.5309 8.4881 8.4881
14.0697 12.9204 11.9847 11.3838 11.3838
15.8758 14.8599 14.1036 13.6785 13.6785
17.4545 16.5397 15.8912 15.5483 15.5483
18.7917 17.9531 17.3734 17.0750 17.0750
19.8882 19.1074 18.5745 18.3034 18.3034
20.7511 20.0139 19.5137 19.2606 19.2606
21.3889 20.6832 20.2058 19.9648 19.9648
21.8094 21.1243 20.6614 20.4280 20.4280
22.0182 21.3433 20.8875 20.6577 20.6577
22.0182 21.3433 20.8875 20.6577 20.6577
fx >>
激活 Windows
转到"设置"以激活 Windows。